Samiul Haque Rudra

Machine Learning Engineer | On-Device AI, NLP, Computer Vision

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Kaggle | Hugging Face

PROFILE

B.Sc. CSE graduate focused on machine learning, with hands-on experience building and evaluating deep learning models for transformer-based NLP, computer vision, ECG/EEG signal classification, and edge deployment. Skilled in PyTorch, TensorFlow, and Scikit-learn, with strong data structures and software engineering fundamentals.

FEATURED THESIS

PrivAdapt: A Privacy-Oriented, Self-Recalibrating On-Device LLM System

Team of six | United International University, CSE | Supervisor: Ms. Sadia Islam, Assistant Professor

Built a framework running 4-bit quantized 1B and 3B language models on a CPU-only laptop. Contributed self-recalibrating routing for distribution shift, configurable privacy levels, and compute-adaptive model switching. Results: 96.0% routing accuracy on 150 queries; 19.8% lower normalized routing cost after drift; 2.23x 1B-tier speedup; no exact raw-query transmission across 450 inspected payloads.

EXPERIENCE & LEADERSHIP

Research Intern — *Brain Computer Interface Research Lab, IRIIC, UIU*

Researched EEG-based BCI systems, signal-processing techniques, AI-driven cognitive analysis, and neuromarketing applications.

President — *UIU Computer Club (UIUCCL)*

Led ML/AI and cybersecurity workshops; mentored students in competitive programming and research.

Outreach & Engagement Lead — *UIU CSE FEST 2025*

Coordinated outreach for participants, sponsors, and partner institutions.

SELECTED PROJECTS

Implicit Toxicity Detection
Context-aware transformer and preprocessing pipeline for noisy Bangla-English code-mixed conversations.

Real-Time Drowsiness Detection
EAR/MAR driver monitoring optimized for Raspberry Pi with adaptive thresholding.

EEG Band Analysis
Multi-channel band-pass filtering and delta/theta/alpha/beta band extraction.

ECG Anomaly Detection
Hybrid Random Forest + LSTM arrhythmia detection with R-peak and morphological feature extraction.

Green AI Inference
Carbon-aware dynamic model switching designed to balance inference efficiency and stability.

Research Resource Platform
MERN platform with bcrypt authentication, file sharing, search, dashboards, and a MongoDB REST API.

SKILLS

Languages	Python, Java, C++, JavaScript, SQL
Machine Learning	Deep Learning, Transformers, NLP, Computer Vision, Time-Series, Model Optimization, Edge Inference
Frameworks	TensorFlow, PyTorch, Scikit-learn, OpenCV, MediaPipe, MNE, EEGLAB, Librosa, Pandas, NumPy
Core CS & Web	Data Structures & Algorithms, OOP, DBMS, Software Engineering, MERN Stack, Git

EDUCATION & ACHIEVEMENT

B.Sc. Computer Science & Engineering
United International University, Dhaka | CGPA 3.68 / 4.0 | Graduated 2026

Champion
Deep Learning in Computational and Spatial Biology - Inter-University Competition